Binomial Distribution

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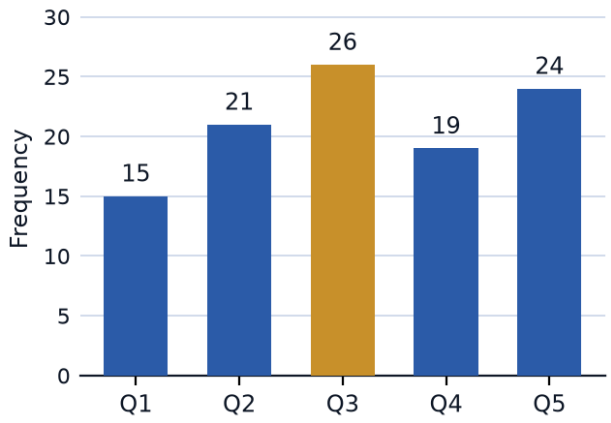
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Conditions for a binomial distribution

State the four conditions required for a random variable to follow a binomial distribution.

2 [Foundational] • Target: Conditions for a binomial distribution

A survey asks 50 people whether they support a new policy, and X is the number who say yes, assuming each person's response is independent with the same probability of support. Explain why X follows a binomial distribution.



3 [Foundational] • Target: Conditions for a binomial distribution

A researcher rolls a die 15 times and counts the number of times a 3 appears. Explain why this count follows a binomial distribution.

4 [Foundational] • Target: Calculating binomial probabilities

A fair coin is tossed 8 times. Find the probability of getting exactly 5 heads, using $P(X = r) = \binom{n}{r} p^r (1-p)^{n-r}$.

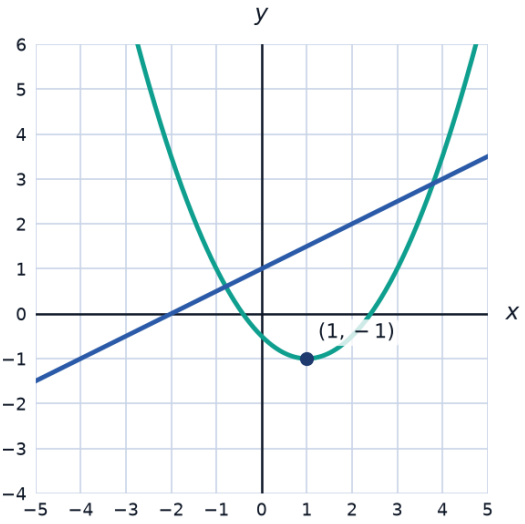
5 [Foundational] • Target: Calculating binomial probabilities

A factory produces items with a defect rate of 10%. In a sample of 6 items, find the probability that exactly 2 are defective.

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6 [Foundational] • Target: Calculating binomial probabilities

A fair coin is tossed 6 times. Find the probability of getting exactly 4 heads.



7 [Foundational] • Target: Calculating binomial probabilities

A factory produces items with a defect rate of 20%. In a sample of 5 items, find the probability that exactly 1 is defective.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Cumulative binomial probabilities

A biased coin has $P(\text{heads}) = 0.3$. It is tossed 5 times. Find $P(X \leq 1)$, where X is the number of heads.

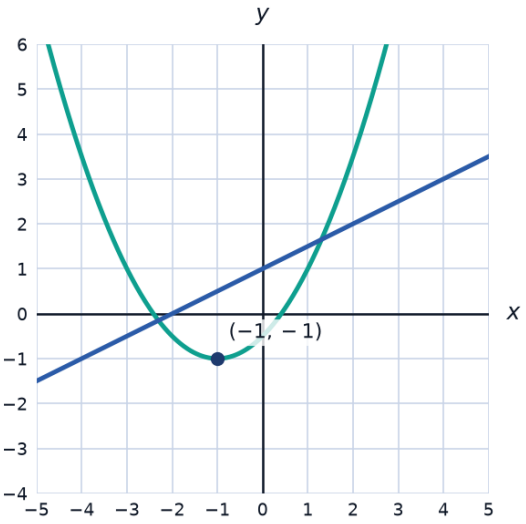
9 [Intermediate] • Target: Cumulative binomial probabilities

Using the same distribution $X \sim B(5, 0.3)$, find $P(X \geq 2)$.

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10 [Intermediate] • Target: Cumulative binomial probabilities

A biased coin has $P(\text{heads}) = 0.2$. It is tossed 4 times. Find $P(X \leq 1)$, where X is the number of heads.



11 [Intermediate] • Target: Mean and variance of a binomial distribution

A random variable $X \sim B(40, 0.25)$. Find the mean and variance, using $E(X) = np$ and $\text{Var}(X) = np(1-p)$.

12 [Intermediate] • Target: Mean and variance of a binomial distribution

A random variable $X \sim B(60, 0.2)$. Find the mean and variance.

13 [Intermediate] • Target: Mean and variance of a binomial distribution

A random variable $X \sim B(80, 0.1)$. Find the mean and variance.

14 [Intermediate] • Target: Mean and variance of a binomial distribution

A random variable $X \sim B(50, 0.6)$. Find the mean and variance.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

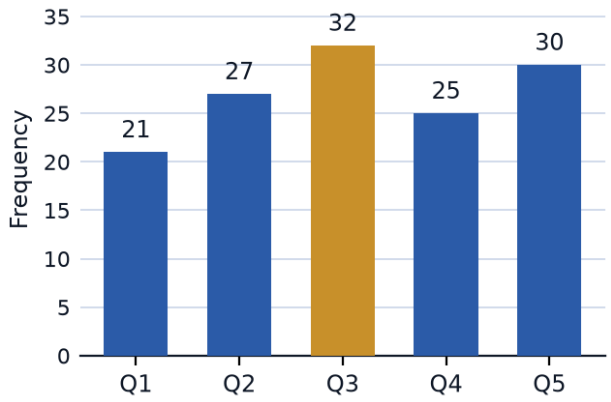
15 [Challenging] • Target: Finding an unknown parameter • Cross-Lesson: 14.01 Energy, Work & Power

A random variable $X \sim B(n, 0.4)$ has mean $E(X) = 20$. Find n.

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16 [Challenging] • Target: Finding an unknown parameter • Cross-Lesson: 15.01 Representation of Data

A random variable $X \sim B(25, p)$ has mean $E(X) = 10$. Find p.



17 [Challenging] • Target: Finding an unknown parameter • Cross-Lesson: 16.01 Permutations & Combinations

A random variable $X \sim B(n, 0.25)$ has variance $\text{Var}(X) = 9$. Find n.

18 [Challenging] • Target: Finding an unknown parameter • Cross-Lesson: 17.01 Discrete Random Variables

A random variable $X \sim B(n, 0.5)$ has variance $\text{Var}(X) = 12.5$. Find n.

19 [Challenging] • Target: Using the binomial distribution to model a real scenario • Cross-Lesson: 19.01 Normal Distribution

A multiple-choice quiz has 10 questions, each with 4 options, only one of which is correct. A student guesses randomly on every question. Find the probability the student gets at least 8 questions correct.

20 [Challenging] • Target: Using the binomial distribution to model a real scenario • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

A basketball player has a 70% free-throw success rate. Find the probability that the player makes exactly 3 out of 5 free throws.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Binomial Criteria & Single Probabilities

Which of the following is NOT a required condition for a variable X to have a binomial distribution $B(n, p)$?

a. The probability of success changes with each trial

b. There is a fixed number of trials n

c. Each trial results in one of only two outcomes (success or failure)

d. Trials are independent of one another

22 [Foundational] • Target: Binomial Criteria & Single Probabilities

If $X \sim B(6, 0.3)$, calculate $P(X = 3)$ correct to 4 decimal places.

a. 0.1852

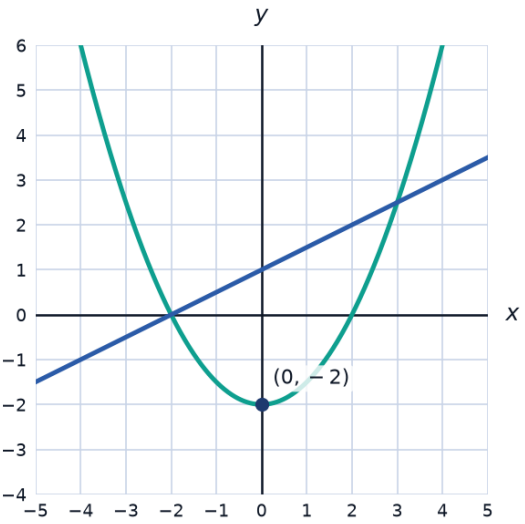
b. 0.2541

c. 0.3087

d. 0.1176

23 [Foundational] • Target: Binomial Criteria & Single Probabilities

If $X \sim B(5, 0.4)$, find $P(X = 0)$.



a. 0.0778

b. 0.0102

c. 0.0625

d. 0.1000

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24 [Foundational] • Target: Cumulative Binomial Probabilities

If $X \sim B(4, 0.5)$, find $P(X \geq 2)$.

a. $\frac{11}{16} = 0.6875$

b. $\frac{1}{2} = 0.5000$

c. $\frac{5}{16} = 0.3125$

d. $\frac{3}{8} = 0.3750$

25 [Foundational] • Target: Cumulative Binomial Probabilities

A manufacturer knows that 5% of lightbulbs are defective. In a random sample of 20 bulbs, what is the probability that at most 1 is defective?

a. 0.7358

b. 0.6824

c. 0.8245

d. 0.9128

26 [Foundational] • Target: Cumulative Binomial Probabilities

A fair coin is tossed 6 times. Find the probability of obtaining at least 5 heads.

a. $\frac{7}{64} \approx 0.1094$

b. $\frac{6}{64} \approx 0.0938$

c. $\frac{1}{64} \approx 0.0156$

d. $\frac{11}{64} \approx 0.1719$

27 [Foundational] • Target: Mean & Variance of the Binomial Distribution

A random variable $X \sim B(48, 0.25)$. Find its mean and standard deviation.

a. Mean $= 12, \sigma = 3$

b. Mean $= 12, \sigma = 9$

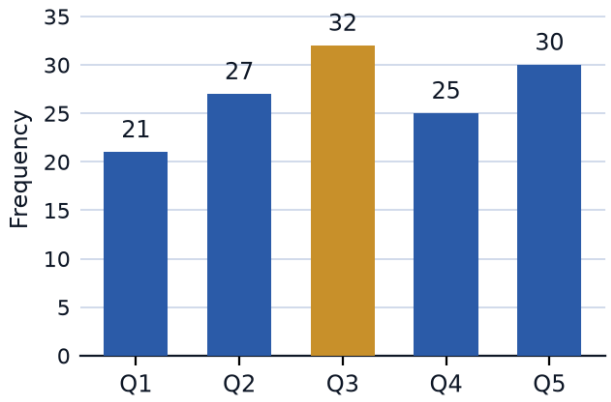
c. Mean $= 16, \sigma = 4$

d. Mean $= 12, \sigma = \sqrt{3}$

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28 [Foundational] • Target: Mean & Variance of the Binomial Distribution

For a binomial distribution $X \sim B(n, p)$, the mean is 6 and the variance is 3.6. Find n and p.



a. $n = 15, p = 0.4$

b. $n = 20, p = 0.3$

c. $n = 12, p = 0.5$

d. $n = 24, p = 0.25$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Mean & Variance of the Binomial Distribution

What is the maximum possible variance of a binomial distribution with $n = 40$ trials?

a. 10

b. 20

c. 5

d. 40

30 [Intermediate] • Target: Contextual Modeling & Problem Solving

A treatment is successful for 80% of patients. If 10 patients are treated, what is the most likely number of successful outcomes (the mode)?

a. 8

b. 7

c. 9

d. 10

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31 [Intermediate] • Target: Contextual Modeling & Problem Solving

How many times must a fair die be rolled so that the probability of getting at least one 6 is greater than 0.9 ?

a. At least 13 rolls

b. At least 10 rolls

c. At least 6 rolls

d. At least 18 rolls

32 [Intermediate] • Target: Contextual Modeling & Problem Solving

In an exam, 70% of candidates pass. Out of 8 randomly chosen candidates, what is the probability that exactly 6 pass, given that at least 5 pass?

a. ≈ 0.380

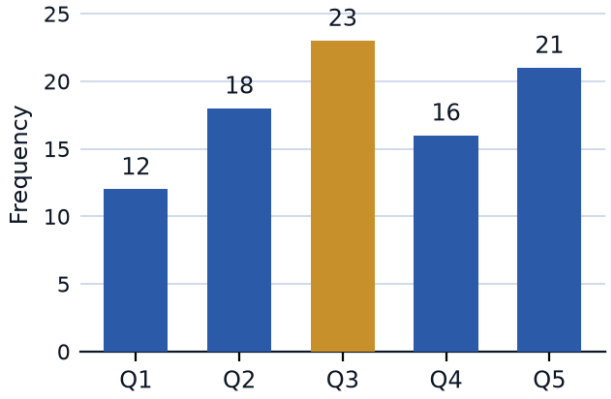
b. ≈ 0.520

c. ≈ 0.296

d. ≈ 0.420

33 [Intermediate] • Target: Conditions for a binomial distribution

A factory produces light bulbs. The probability that a randomly selected light bulb is defective is p. The number of defective light bulbs in a random sample of n bulbs follows a binomial distribution $X \sim B(n, p)$. It is known that the mean number of defective bulbs in a sample of $n = 25$ is 2. \\ (i) Calculate the value of p. \\ (ii) For a sample of $n = 50$ bulbs, calculate the probability of finding exactly k defective bulbs, where k is the smallest integer such that $P(X \leq k) > 0.9$. \\ (iii) If a sample of $n = 50$ bulbs is taken, find the smallest value of n' such that the probability of observing at least one defective bulb is greater than 0.999 .



a. 0.04

b. 0.08

c. 0.12

d. 0.16

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34 [Intermediate] • Target: Calculating binomial probabilities

The scores on a standardized test are normally distributed with a mean of 100 and a standard deviation of 15. A new scoring algorithm is implemented, and the scores are transformed using the rule $Y = aX + b$, where X is the original score and Y is the new score. The new distribution of scores, Y, also follows a normal distribution with a mean of 120 and a standard deviation of 22.5. Calculate the value of a and the value of b.

a. a = 1.5, b = -30

b. a = -1.5, b = 270

c. a = 1.5, b = 30

d. a = 0.67, b = 53

35 [Intermediate] • Target: Cumulative binomial probabilities

A fair six-sided die is rolled 4 times. Let X be the number of times a 6 is obtained. The possible values for X are 0, 1, 2, 3, and 4. The probability of obtaining a 6 on any given roll is $p = \frac{1}{6}$. The number of trials is $n = 4$. This scenario can be modelled by a binomial distribution, denoted as $X \sim B(n, p)$. Which of the following represents the correct parameters for this binomial distribution?

a. $B(4, 1/6)$

b. $B(6, 1/4)$

c. $B(1, 4/6)$

d. $B(4, 4/6)$

36 [Intermediate] • Target: Mean and variance of a binomial distribution

Cards are dealt from a single shuffled deck without replacement; X is the number of aces dealt. Is X binomially distributed?

a. No — trials are not independent / probability is not constant

b. Yes — but only if $n > 30$

c. Yes — fixed, independent trials with constant success probability

d. No — the number of trials is not fixed

37 [Intermediate] • Target: Finding an unknown parameter

A fair coin is tossed 3 times. Let X be the number of heads obtained. What is the probability that X = 2?

a. $\frac{1}{8}$

b. $\frac{3}{8}$

c. $\frac{1}{2}$

d. $\frac{7}{8}$

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging] • Target: Using the binomial distribution to model a real scenario • Cross-Lesson: 17.01 Discrete Random Variables**

The weights of bags of sugar are normally distributed with a mean of 1000 g and a standard deviation of 5 g. Let X be the weight of a randomly selected bag of sugar. Calculate the probability that a bag of sugar weighs between 990 g and 1010 g.

a. 0.9545

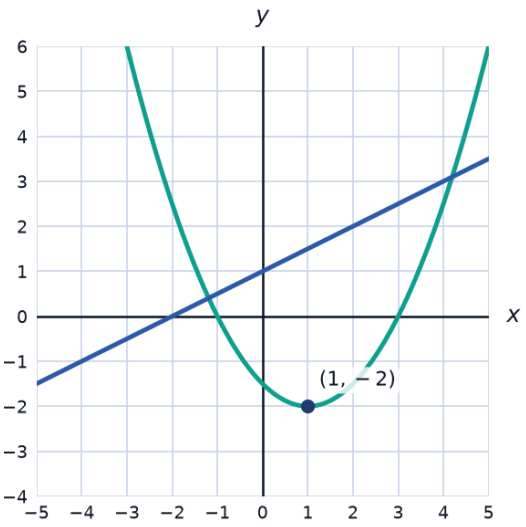
b. 0.6827

c. 0.9772

d. 0.0228

39 **[Challenging] • Target: Distinguishing binomial situations from non-binomial situations • Cross-Lesson: 19.01 Normal Distribution**

A factory tests 30 independent components, each with a fixed probability of being defective; X is the number of defective components. Is X binomially distributed?



a. Yes — fixed, independent trials with constant success probability

b. No — the number of trials is not fixed

c. No — trials are not independent / probability is not constant

d. Yes — but only if $n > 30$

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40 [Challenging] • Target: A well-designed word problem: quality control sampling • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

A die is rolled 15 times; X is the number of sixes. Is X binomially distributed?

a. No — trials are not independent / probability is not constant

b. Yes — but only if $n > 30$

c. Yes — fixed, independent trials with constant success probability

d. No — the number of trials is not fixed

41 [Challenging] • Target: Binomial Criteria & Single Probabilities • Cross-Lesson: 3.01 Graphs & Transformations

A fair coin is tossed 20 times; X is the number of heads. Is X binomially distributed?

a. No — trials are not independent / probability is not constant

b. Yes — but only if $n > 30$

c. Yes — fixed, independent trials with constant success probability

d. No — the number of trials is not fixed

42 [Challenging] • Target: Cumulative Binomial Probabilities • Cross-Lesson: 4.01 Coordinate Geometry

A survey asks 50 randomly and independently selected people the same yes/no question; X is the number who say yes. Is X binomially distributed?

a. No — trials are not independent / probability is not constant

b. No — the number of trials is not fixed

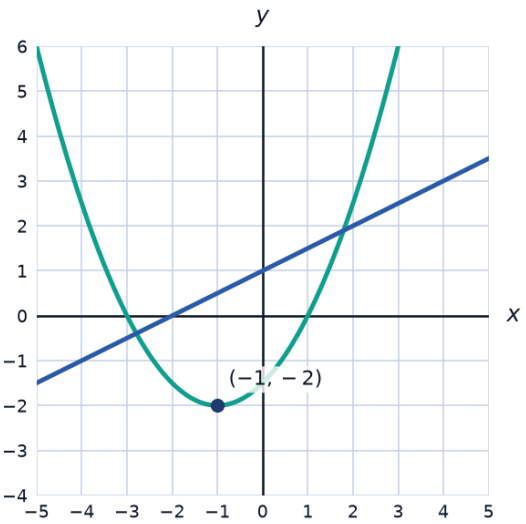
c. Yes — fixed, independent trials with constant success probability

d. Yes — but only if $n > 30$

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43 [Challenging] • Target: Mean & Variance of the Binomial Distribution • Cross-Lesson: 5.01 Trigonometry

Given $X \sim B(73, 17/20)$, find $E(X)$.



a. 1221/20

b. 1251/20

c. 1261/20

d. 1241/20

44 [Challenging] • Target: Contextual Modeling & Problem Solving • Cross-Lesson: 6.01 Differentiation & Integration

Given $X \sim B(39, 11/20)$, find $E(X)$.

a. 409/20

b. 429/20

c. 439/20

d. 449/20

45 [Challenging] • Target: Conditions for a binomial distribution • Cross-Lesson: 7.01 Statistics & Probability

Given $X \sim B(38, 13/20)$, find $E(X)$.

a. 247/10

b. 257/10

c. 237/10

d. 126/5

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